

BJT

* BIPOLAR JUNCTION TRANSISTOR (BJT)

BJT is three terminal three layer semiconductor device. The terminals are Emitter Base and collector.

Notes By:-

Prof. Amey P. Raut

CONSTRUCTION:-

The transistor is a three-layer semiconductor device consisting of either two n-type and one p-type layers of material or two p-type and one n-type layers of material.

The former is called an npn transistor and the latter is called a pnp transistor.

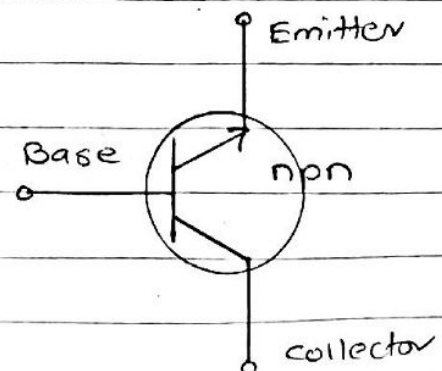
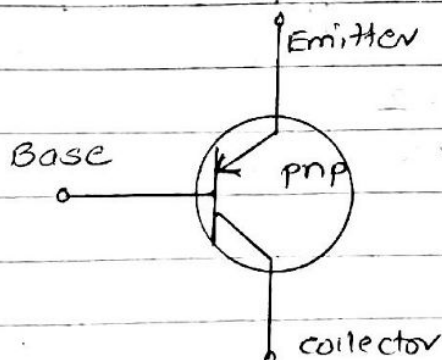
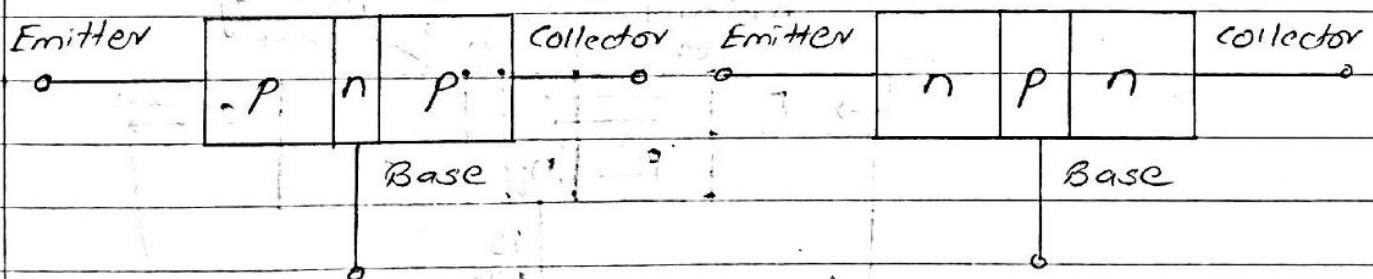


Fig 1: Structures and Symbols.

The emitter layer is heavily doped, the base lightly doped, and the collector only slightly doped.

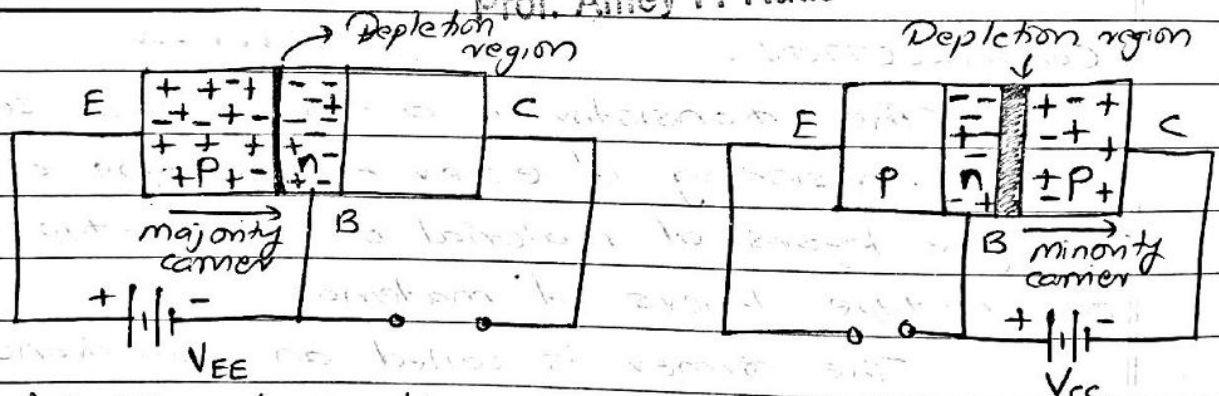
The outer layers have widths much greater than the sandwiched p- or n-type material.

The lower doping levels decreases the conductivity (increases the resistance) of the base material by limiting the number of "free" carriers.

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OPERATION :-



(a) Emitter base bias

(b) Collector base bias

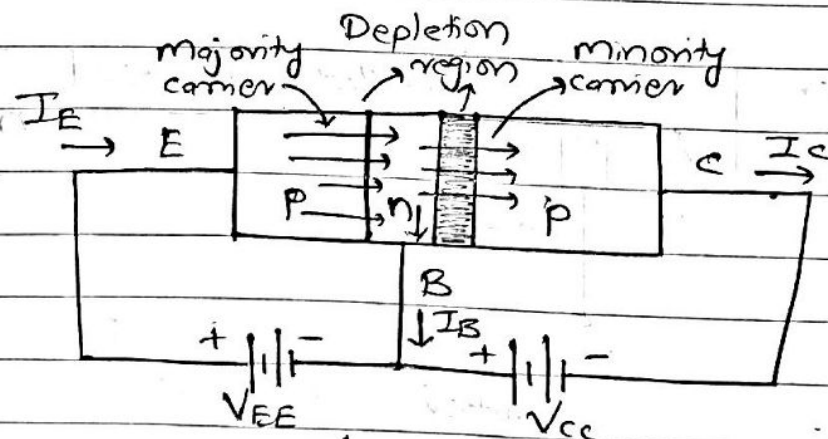


fig :- 2

(c) Biased p-n-p transistor

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In fig 2(a) the pnp transistor has been redrawn without the base-to-collector bias.

Emitter-base junction acts as a forward-biased diode. The depletion region has been reduced in width due to the applied bias, resulting in a heavy flow of majority carriers from the p- to the n-type material.

In fig 2(b) the pnp transistor has been redrawn without the base-to-emitter bias.

collector-base junction acts as a reverse biased. Flow of majority carriers is zero, resulting in only a minority-carrier flow.

* Therefore, one p-n junction of a transistor is reverse-biased, whereas the other is forward-biased.

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In fig 2(c) both biasing potentials have been applied to a p-n-p transistor.

A large number of majority carriers will diffuse across the forward biased p-n junction into the n-type material.

Since the sandwiched n-type material is very thin and has a low conductivity, a very small number of these carriers will take this path of high resistance to the base terminal.

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The larger number of these majority carriers will diffuse across the reverse-biased junction into the p-type material connected to the collector terminal.

Majority carriers can cross the reverse-biased junction with relative ease as that for the reverse

biased diode the injected majority carriers will appear as minority carriers in the n-type material.

Emitter current is the sum of the collector and base currents.

$$I_E = I_C + I_B$$

The collector current comprises two components the majority and minority carriers.

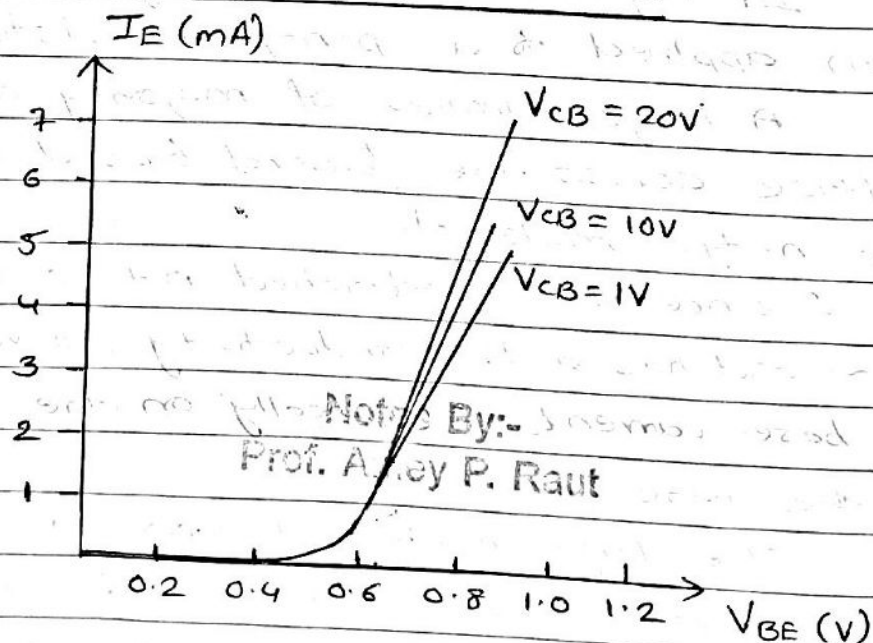
$$I_C = I_{C\text{majority}} + I_{C\text{minority}}$$

The minority-current component is called the leakage current and is given the symbol I_{CO} .

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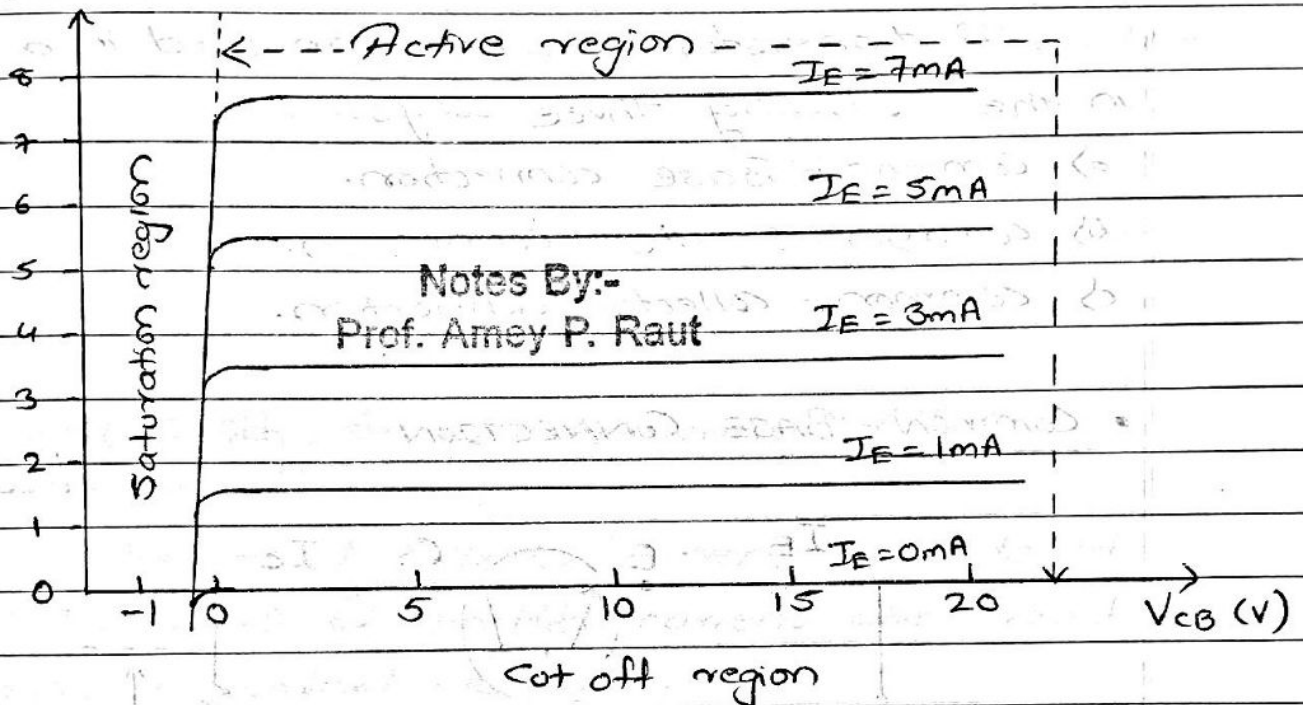
CHARACTERISTICS :- Prof. Arney P. Raut

• Driving point or Input parameters.



Output parameters

I_C (mA)



In the active region the base-emitter junction is forward-biased, whereas the collector-base junction is reverse biased.

In the cut off region the base-emitter and collector-base junctions of a transistor are both reverse-biased.

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In the saturation region the base-emitter and collector-base junctions are forward-biased.

* TRANSISTOR CONNECTIONS :-

A transistor can be connected in a circuit in the following three ways:-

- Common - Base connection.
- Common - Emitter connection.
- Common - collector connection.

• COMMON - BASE CONNECTION

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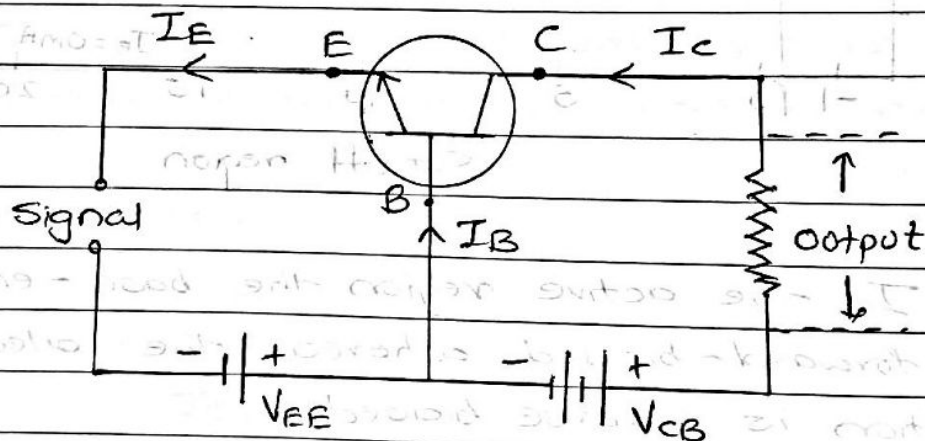


fig 3

In CB circuit arrangement, input is applied between emitter and base and output is taken from collector and base.

Base of the transistor is common to both input and output circuits and hence the name common base connection.

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A common base npn transistor circuit is shown in fig 3.

- Current amplification factor (α).

It is the ratio of output current to input current.

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$$\therefore \alpha = \frac{\Delta I_C}{\Delta I_E} \text{ at constant } V_{CB}$$

- Expression for collector current (I_C)

The total collector current consists of

i) The part of emitter current which reaches the collector terminal i.e. αI_E .

ii) The leakage current I_{leakage} . This current is due to the movement of minority carriers across base-collector junction as it is reverse biased.

$$\therefore I_C = \alpha I_E + I_{\text{leakage}}$$

$$\text{If } I_E = 0 \text{ then } I_{\text{leakage}} = I_{CBO}$$

$$\therefore I_C = \alpha I_E + I_{CBO}$$

$$\therefore I_E = I_C + I_B$$

$$I_C = \alpha (I_C + I_B) + I_{CBO}$$

$$I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

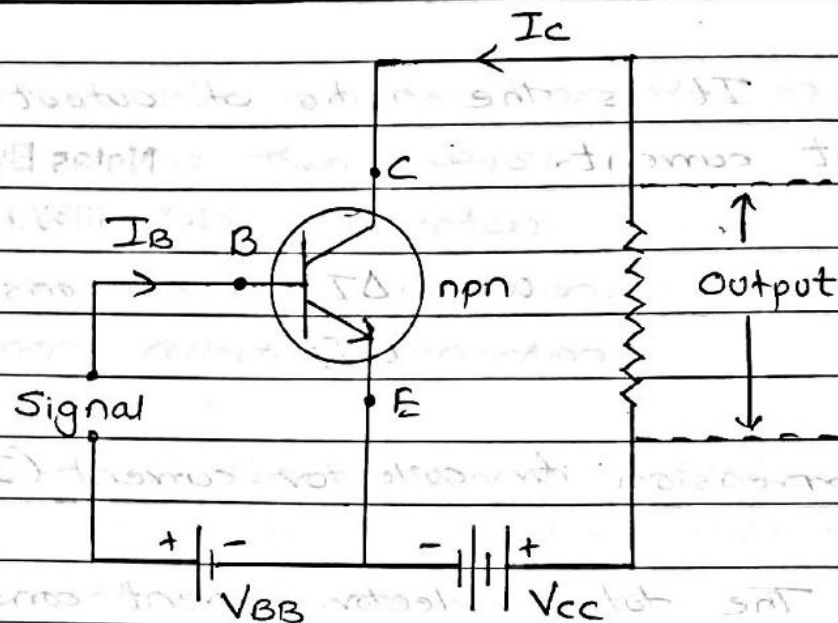
$$\therefore I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha}$$

Thus collector current can be controlled by either the emitter or base current.

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• Common - Emitter Connection



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Fig 4. Prof. Amey P. Raut

In CE circuit arrangement, input is applied between base and emitter and output is taken from the collector and emitter.

Emitter of the transistor is common to both input and output circuits and hence the name common emitter connection.

A common emitter npn transistor circuit is shown in Fig 3.

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• Base current amplification factor (β)

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

• Relation between β and α .

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

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We have $I_E = I_B + I_C$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

$$\Delta I_B = \Delta I_E - \Delta I_C$$

$$\therefore \beta = \frac{\Delta I_C}{\Delta I_E - \Delta I_C}$$

Dividing by ΔI_E we have

$$\beta = \frac{\Delta I_C / \Delta I_E}{1 - \Delta I_C / \Delta I_E}$$

But $\frac{\Delta I_C}{\Delta I_E} = \alpha$

$$\therefore \beta = \frac{\alpha}{1 - \alpha}$$

• Expression for collector current (I_C)

We have

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In exams
derive I_C

$$I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha}$$

If $I_B = 0$ then the circuit is called as

$$\frac{I_{CBO}}{1-\alpha} = I_{CEO}, \text{ collector-emitter current with base open}$$

$$\therefore I_C = \frac{\alpha I_E + I_{CEO}}{1-\alpha}$$

$$I_C - \alpha I_C = I_{CEO}$$

$$\therefore I_C = \beta I_B + I_{CEO}$$

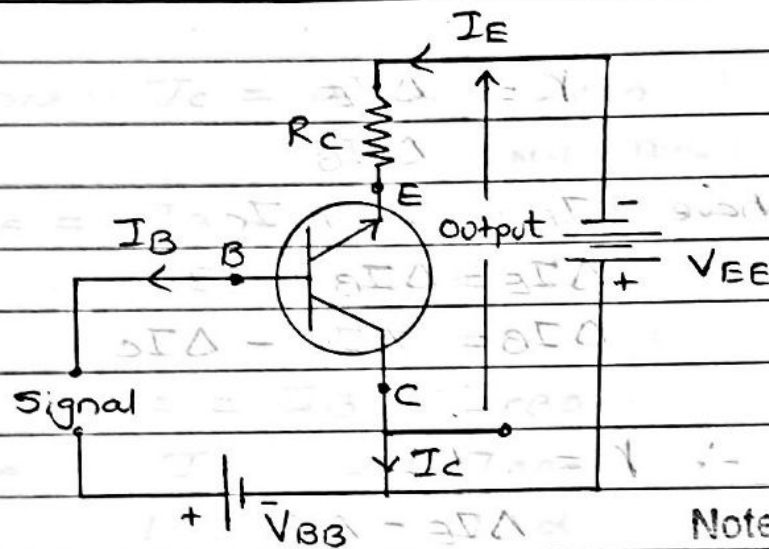
Significance of β Notes By:-
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$$\beta = \frac{\alpha}{1-\alpha}$$

$\therefore$ as α approaches unity β approaches infinity.

$\therefore$ The current gain in common emitter connection is very high.

Common - Collector Connection



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Fig 5

In CC circuit arrangement, input is applied between base and collector and output is taken from the collector and emitter.

Collector of the transistor is common to both input and output circuits and hence the name common - collector connection.

A common-collector npn transistor circuit is shown in Fig 5.

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• Current amplification factor (β)

$$\beta = \frac{\Delta I_E}{\Delta I_B}$$

• Relation between r and α

$$r = \frac{\Delta I_E}{\Delta I_B}$$

We have $I_E = I_B + I_C$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

$$\Delta I_B = \Delta I_E - \Delta I_C$$

$$\therefore r = \frac{\Delta I_E}{\Delta I_E - \Delta I_C}$$

Dividing by ΔI_E Notes By:-
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$$r = \frac{1}{1 - \Delta I_C / \Delta I_E}$$

Let $\alpha = \Delta I_C / \Delta I_E$ (ratio of collector current change to emitter current change)

$$\therefore r = \frac{1}{1 - \alpha} \quad \left(\because \alpha = \frac{\Delta I_C}{\Delta I_E} \right)$$

• Relation between r and β

$$\text{we have } \beta = \frac{\alpha}{1 - \alpha}$$

(1) $\alpha = 1 - \alpha$ Notes By:-

adding 1 both sides Prof. Amey P. Raut

$$\beta + 1 = \frac{\alpha}{1 - \alpha} + 1 = \frac{\alpha + 1 - \alpha}{1 - \alpha}$$

$$\therefore \beta + 1 = \frac{1}{1 - \alpha} = r$$

$$\therefore (\beta + 1) = r$$

• Expression for collector current (I_c) :- *

we know $I_c = \alpha I_E + I_{CBO}$

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$I_E = I_B + I_c$ Prof. Amey P. Raut

$I_E = I_B + \alpha I_E + I_{CBO}$

$(1 - \alpha) I_E = I_B + I_{CBO}$

$I_E = \frac{I_B + I_{CBO}}{1 - \alpha}$

$\therefore I_E = \gamma I_B + \gamma I_{CBO} = I$

$\therefore I_c \approx I_E = (\beta + 1) I_B + (\beta + 1) I_{CBO}$

The common collector circuit has very high input resistance and very low output resistance. Due to this reason, the voltage gain provided by the circuit is always less than 1.

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$(1) = \frac{1}{\beta + 1}$

* LEAKAGE CURRENT IN TRANSISTOR:-

Leakage current is because of minority carriers present in semiconductor devices. Since minority carriers are dependent on temperature the leakage current is also temperature sensitive and hence leakage current is not controllable.

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For common-base circuit we have

$$I_c = \alpha I_E + I_{CBO}$$

where I_{CBO} is the leakage current which flows through the circuit when $I_E = 0$

Similarly for common-emitter circuit we have

$$I_c = \beta I_B + \frac{I_{CBO}}{1-\alpha}$$

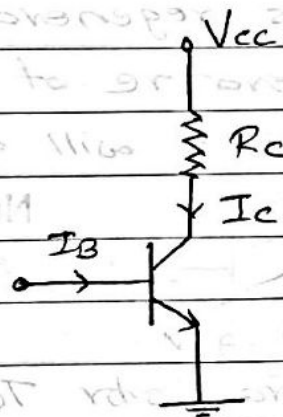
$$\therefore I_c = \beta I_B + (1+\beta) I_{CBO}$$

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$$\left(\because \frac{1}{1-\alpha} = (1+\beta) \right)$$

* THERMAL RUNAWAY:-

The process of self destruction of transistor because of excessive heat is called as thermal Runaway.



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Fig 6

For above transistor amplifier, the temperature sensitive collector current is given by

$$I_c = \beta I_B + (1 + \beta) I_{CBO}$$

Now if ambient temperature increases the minority carriers in transistor increases which gives rise to leakage current.

$$\text{Now } I_c \propto (1 + \beta) I_{CBO}$$

$\therefore$ As the leakage current increases, the collector current rises.

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$$\text{we have Transistor heat} = I_c^2 \times R_t$$

Therefore as I_c increases the heat across transistor increases.

Now when heat of transistor increases it increases the temperature of transistor. more temperature will give rise to more minority carriers which will increase leakage current through transistor. As a result, collector current increases. Thus, this regenerative process will increase the temperature of transistor beyond its extreme limit which will destroy the transistor.

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Ambient Temp

Transistor Temp ↑

$$\text{Heat} = I_C^2 R_t \uparrow$$

minority carriers ↑

 $I_C \uparrow$ $I_{CBO} \uparrow$

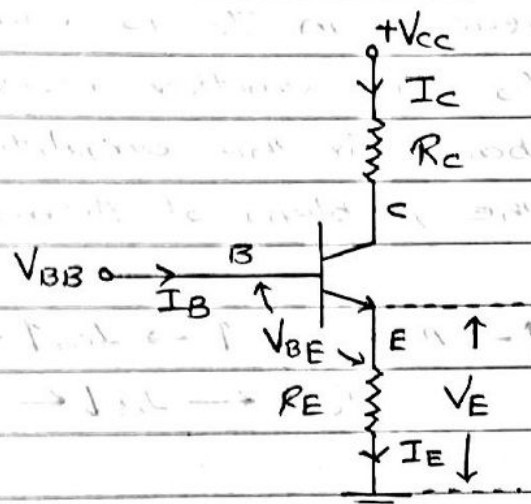
Thus, thermal Runaway is a regenerative destructive process of BJT.

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* Process to avoid Thermal Runaway

The process of thermal runaway can be terminated by inserting resistance between emitter and ground. This is also called self biasing of transistor.



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Fig 7

Now, if ambient temperature increases heat of transistor increases and therefore minority increases. This will give rise to leakage current.

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$$I_c = \beta I_B + (1 + \beta) I_{co}$$

From above eqn, if I_{co} increases the collector current I_c will increase which will increase the emitter current I_E . The emitter current I_E flows through R_E which would produce a drop across it i.e. V_E will increase.

$$V_{BE} = V_B - V_E$$

From above eqn it can be seen that as V_E will increase, V_{BE} will decrease which will

further decrease I_B because

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$I_B \propto V_{BE}$ Prof. Amey P. Raut

As I_B decreases, I_C also decreases ($I_C = \beta I_B$)

Thus increases in I_C is compensated by decrease in I_B . The emitter resistor R_E provides negative feedback for the amplifier which helps to eliminate the problem of Thermal Runaway.

Ambient Temp $\uparrow \rightarrow$ Minority $\uparrow \rightarrow I_{CO} \uparrow \rightarrow I_C \uparrow \rightarrow I_E \uparrow$
 $I_C \downarrow \leftarrow I_B \downarrow \leftarrow V_{BE} \downarrow \leftarrow V_E \uparrow \leftarrow$

* NEED OF BIASING

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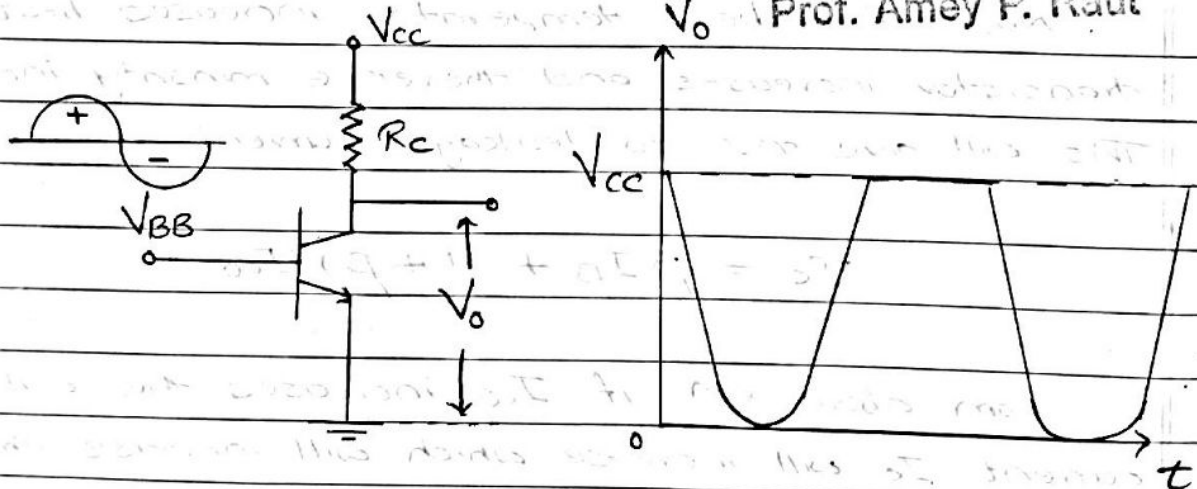


Fig 8

Equation of o/p voltage is given by

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CC} - I_C R_C - V_O = 0$$

$$V_O = V_{CC} - I_C R_C$$

when $V_{in} = 0$, $I_B = 0$, $I_C = \beta I_B = 0$

$$\therefore V_o = -V_{cc}$$

As we increase V_{in} , the I_B increases hence I_C .

$$V_o = V_{cc} - I_C R_C \quad I_C \uparrow$$

Output voltage decreases.

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when $V_{in(max)}$, $I_{B(max)}$, $I_{C(max)} \therefore V_o \approx 0$

when V_{in} is negative, transistor does not conduct

hence $I_B = 0$, $I_C = 0$, $I_C R_C = 0 \therefore V_o = V_{cc}$

Thus for negative input, the o/p is V_{cc} .

Conclusion:-

(i) O/p is 180° phase shifted and amplified.

(ii) It does not provide amplification of negative half cycle of input.

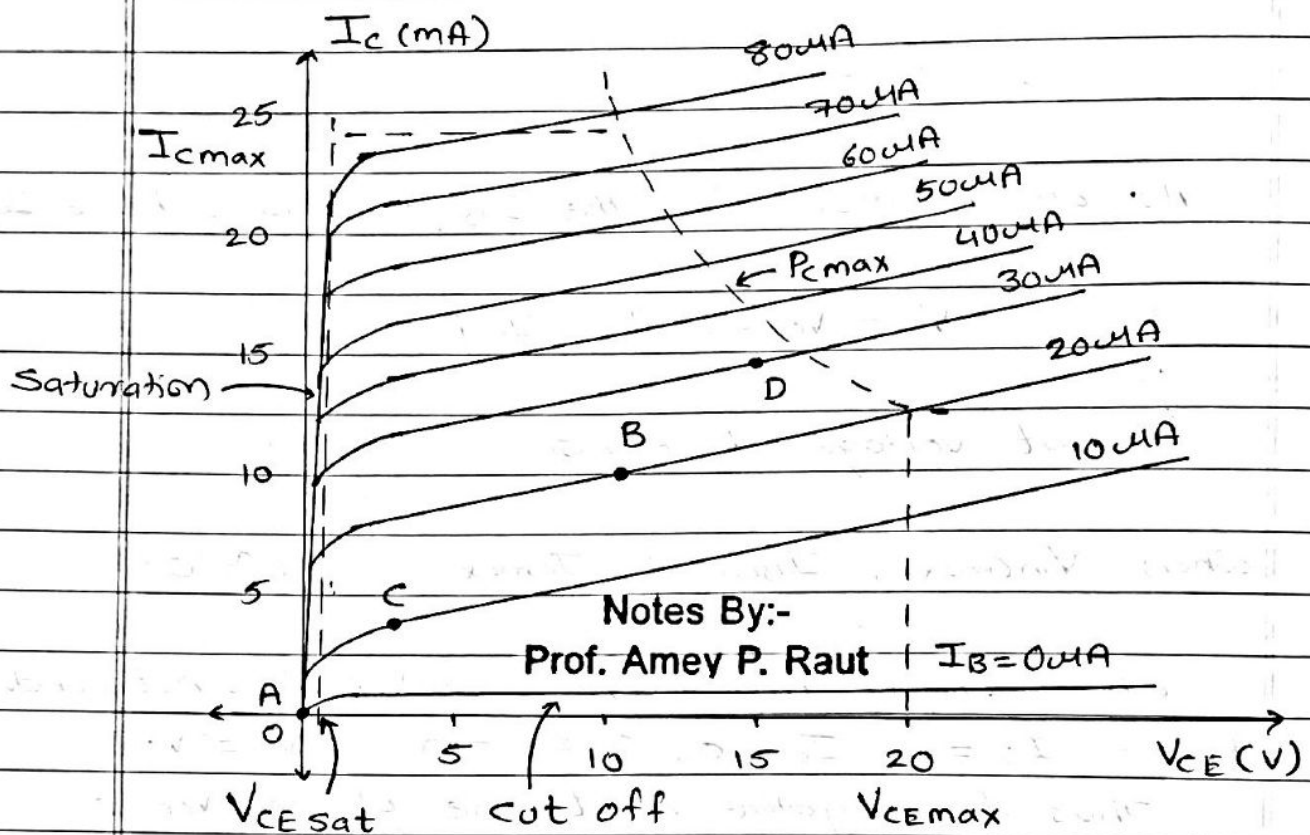
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The transistor does not conduct for negative half cycle of i/p. For conduction of positive as well as negative half cycle the input signal needs to be shifted. This is done through biasing.

Biasing is a process in which input signal is brought above zero level.

Biasing is accomplished by inserting positive battery in series with input signal.

* OPERATING POINT

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Fig. 9

Biasing means application of dc voltage to establish a fixed level of current and voltages.

For transistor amplifiers the resulting dc current and voltage establish an operating point on the characteristics that define the region that will be employed for amplification of the applied signal. This point is also called as quiescent point.

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Figure 9 shows a general output characteristics with four operating points indicated within the active region.

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The maximum ratings are indicated by I_{Cmax} → collector current, V_{CEmax} collector-emitter v_{tg}. and maximum power P_{max} is defined by the curve cut off region is defined by $I_B \leq 0.4A$ and saturation region by $V_{CE} \leq V_{CEsat}$.

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If no bias were used, the device would initially be completely off, resulting in a Q-point at A, zero current through and zero voltage across the device. Thus point A is not suitable.

For point B, if a signal is applied to the circuit the device will vary in current and voltage from the Q point, allowing the device to react to both positive and negative cycles of the input signal. The voltage and current of the device will vary, but not enough to drive the device into cut-off or saturation. Also point B is a point of linear operation as spacing between I_B is linear.

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Point C would allow some positive and negative variation of the output signal, but the peak to peak value would be limited by the proximity of $V_{CE} = 0V$ and $I_C = 0.4A$, point D also has a lot of nonlinearities.

Point D sets the device Q point near the maximum voltage and power level. The output voltage swing in the positive direction is thus limited if the maximum voltage is not to be exceeded.

Point B therefore seems the best point in terms of linear gain and largest possible voltage and current swing, desired condition for small-signal amplifiers.

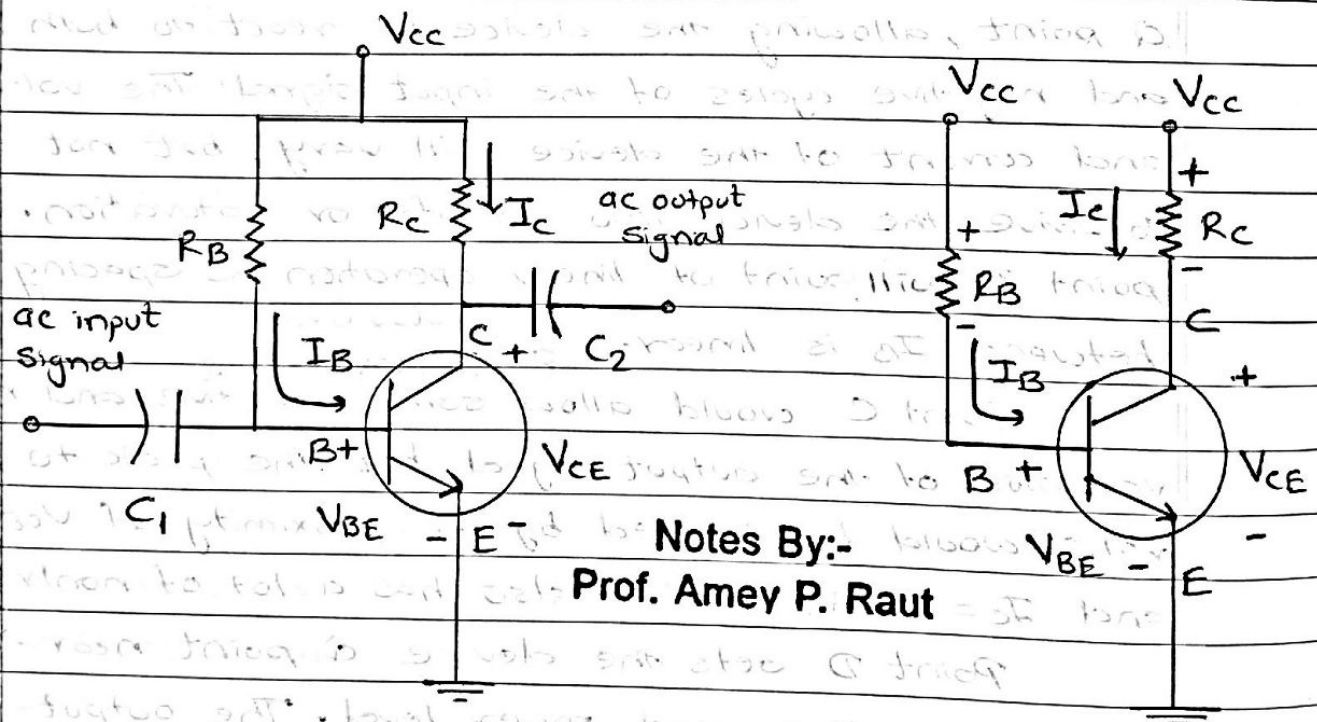
Biasing of BJT outside the maximum limits would cause shortening of the lifetime of the device or destruction of the device.

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Q point is also temperature dependent, thus the network design must provide a degree of temperature stability so that temperature changes result in minimum changes in the Q point. This maintenance of the Q point can be specified by a stability factor (S).

* FIXED BIAS CONFIGURATION



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Fig 10 Fixed Biased circuit Fig 11 DC Equivalent

The fixed-bias circuit is the simplest transistor dc bias configuration. Fig 10 shows fixed bias circuit employed using npn transistor. Same can be employed using pnp transistor by reversing the current and voltage polarities.

For DC analysis the network is isolated from the indicated ac levels by replacing the capacitors with open-circuit equivalent.

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The dc supply V_{CC} can be separated into two supplies to permit a separation of input and output circuits.

→ Forward Bias of Base-Emitter

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Consider first the base-emitter circuit loop.

Applying KVL equations in the clockwise direction for the loop, we obtain-

$$+V_{CC} - I_B R_B - V_{BE} = 0$$

Solving the equation for the current I_B , we get

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \quad \dots \dots \dots (1)$$

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The voltage across R_B is the applied voltage V_{CC} at one end, less the drop across the base to emitter junction (V_{BE})

The supply voltage V_{CC} and the base-emitter voltage V_{BE} are constant, the selection of a base resistor R_B sets the level of base current for the operating point.

→ Collector - Emitter Loop:-

The collector-emitter section of the network appears in Fig 11 with the indicated direction of current I_C .

The magnitude of the collector current is related directly to I_B

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$$I_C = \beta I_B \quad (2)$$

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From equation (1) we can state that, base current is controlled by the level of R_B and from eqn (2) I_C is related to I_B by a constant β , the magnitude of I_C is not a function of the resistance R_C . Changing R_C to any level will not affect the level of I_B or I_C as long as transistor is operated in active region.

Applying KVL in the clockwise direction to the output loop results in the following.

$$V_{CE} + I_C R_C - V_{CC} = 0$$

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$$\therefore V_{CE} = V_{CC} - R_C I_C \quad (3)$$

Thus the voltage across the collector-emitter region of a transistor in the fixed-bias configuration is the supply voltage less the drop across R_C .

Also from fig 11 we can state

$$V_{CE} = V_C - V_E \quad (4)$$

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$V_{CE} \rightarrow$ voltage from collector to emitter.

$V_C \rightarrow$ voltage from collector to ground.

$V_E \rightarrow$ voltage from emitter to ground.

In fixed bias $V_E = 0V$

$$\therefore V_{CE} = V_C = \dots (5)$$

Also,

$$V_{BE} = V_B - V_E \quad (6)$$

$$\therefore V_{BE} = V_B \quad (7)$$

$\rightarrow$ TRANSISTOR SATURATION

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The term saturation is applied to any system where levels have reached their maximum values. For a transistor operating in the saturation region, the current is a maximum value for the particular design.

Saturation conditions are normally avoided because the base-collector junction is no longer reverse-biased and the output amplified signal will be distorted.

In fig 9 saturation region is where the characteristic curves join and the collector-to-emitter voltages is at or below V_{CEsat} . The collector current is relatively high on the characteristics.

Approximating the curves in saturation region in such a way that V_{CE} is assumed to be 0V, the current is relatively high, by applying Ohm's law we can determine the resistance between collector and emitter terminals as follows:

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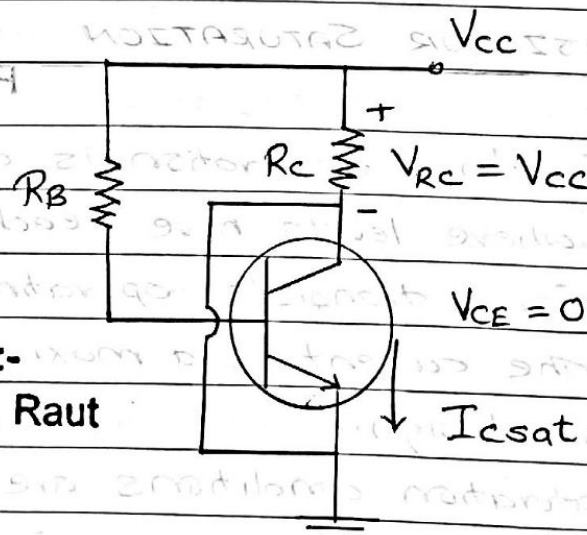
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$$R_{CE} = \frac{V_{CE} \approx 0}{I_C} = 0\Omega$$

To determine the approximate maximum collector current, insert a short-circuit equivalent between collector and emitter of the transistor and calculate the resulting collector current.

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Fig. 12

Saturation current for fixed bias configuration is

$$I_{csat} = \frac{V_{cc}}{R_c} \dots (8)$$

→ Load Line Analysis

For the fixed-bias configuration the load resistor R_C will define the slope of the network equation and resulting intersection between the equation and characteristics of the device.

The smaller the load resistance, the steeper the slope of the network load line.

For forward bias I_C and V_{CE} are related as

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$$V_{CE} = V_{CC} - I_C R_C \quad \dots \dots (9)$$

If we choose I_C to be 0 mA we get,

$$V_{CE} = V_{CC} \mid I_C = 0 \text{ mA} \quad \dots \dots (10)$$

Similarly if we choose $V_{CE} = 0 \text{ V}$ in eqn (9) we get,

$$0 = V_{CC} - I_C R_C$$

$$\therefore I_C = \frac{V_{CC}}{R_C} \mid V_{CE} = 0 \text{ V} \quad \dots \dots (11)$$

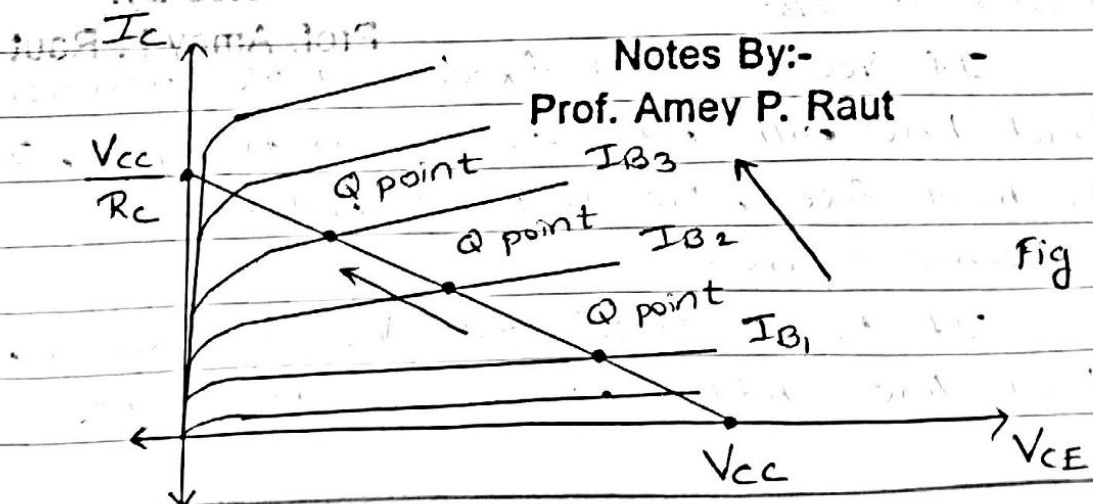


Fig 13

Fig 13 shows the characteristics of a transistor with line drawn from points of equation 10 & 11.

The resulting line on the graph of Fig 13 is called the load line because it is defined by the load resistor R_C .

By solving for the resulting level of I_B we can establish the actual Q-point as shown in Fig 13.

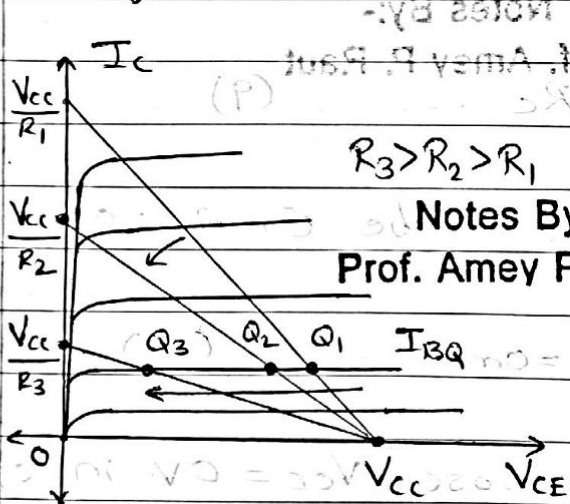


Fig 14

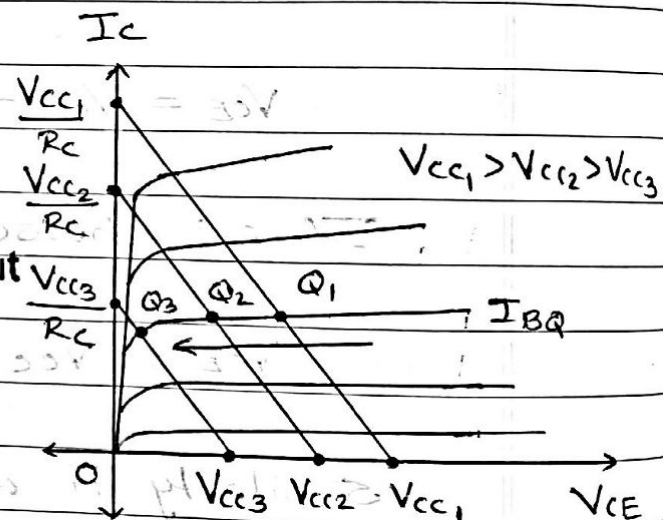


Fig 15

If the level of I_B is changed by varying the value of R_B , the Q point moves up or down the load line as shown in Fig 13 for increasing values of I_B .

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If V_{CC} is held fixed and R_C increased, the load line will shift as shown in Fig 14. If I_B is held fixed, the Q point will move as shown in same figure.

If R_C is fixed and V_{CC} decreased, the load line shifts as shown in Fig 15.

* VOLTAGE-DIVIDER BIAS CONFIGURATION

In the fixed bias configuration the bias current I_{CQ} and voltage V_{CEQ} were a function of the current gain β of the transistor.

Because β is temperature sensitive, especially for silicon transistors, and the actual value of β is usually not well defined, it would be desirable to develop a bias circuit that is less dependent on, or independent of the transistor β .

The voltage divider bias configuration shown in Fig 16 is a network, if analyzed on an exact basis, the sensitivity to changes in β is quite small.

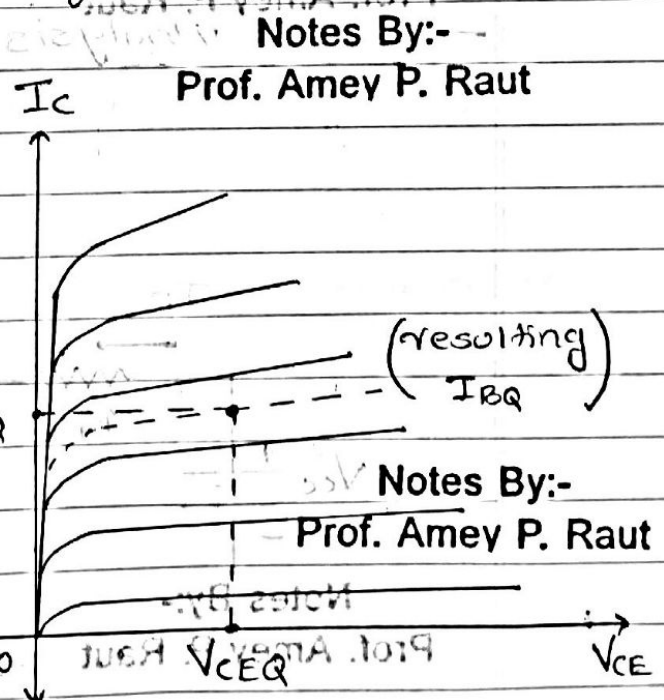
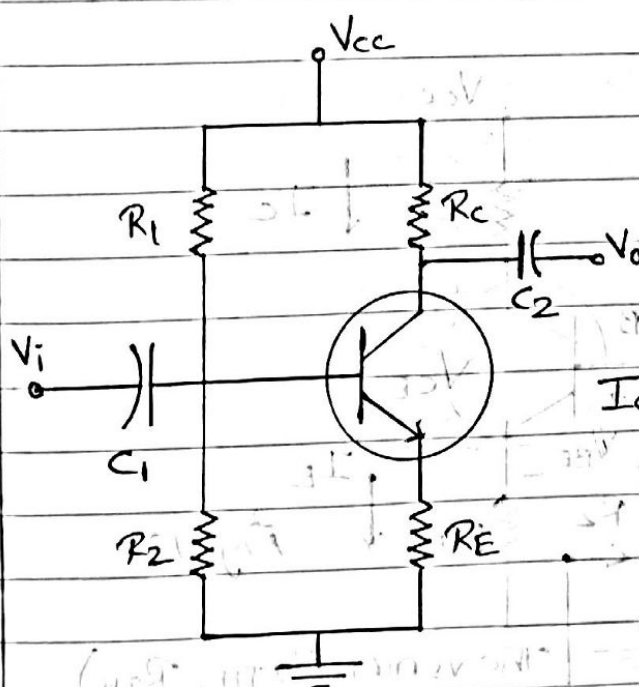


Fig 16

Fig 17

If the circuit parameters are properly chosen the resulting levels of I_{CQ} and V_{CEQ} can be made totally independent of the β .

Q-point is defined by a fixed level of I_{CQ} and V_{CEQ} as shown in fig 17. The level of I_{BQ} will change with the change in β , but the operating point on the characteristics defined by I_{CQ} and V_{CEQ} can remain fixed if the proper circuit parameters are employed.

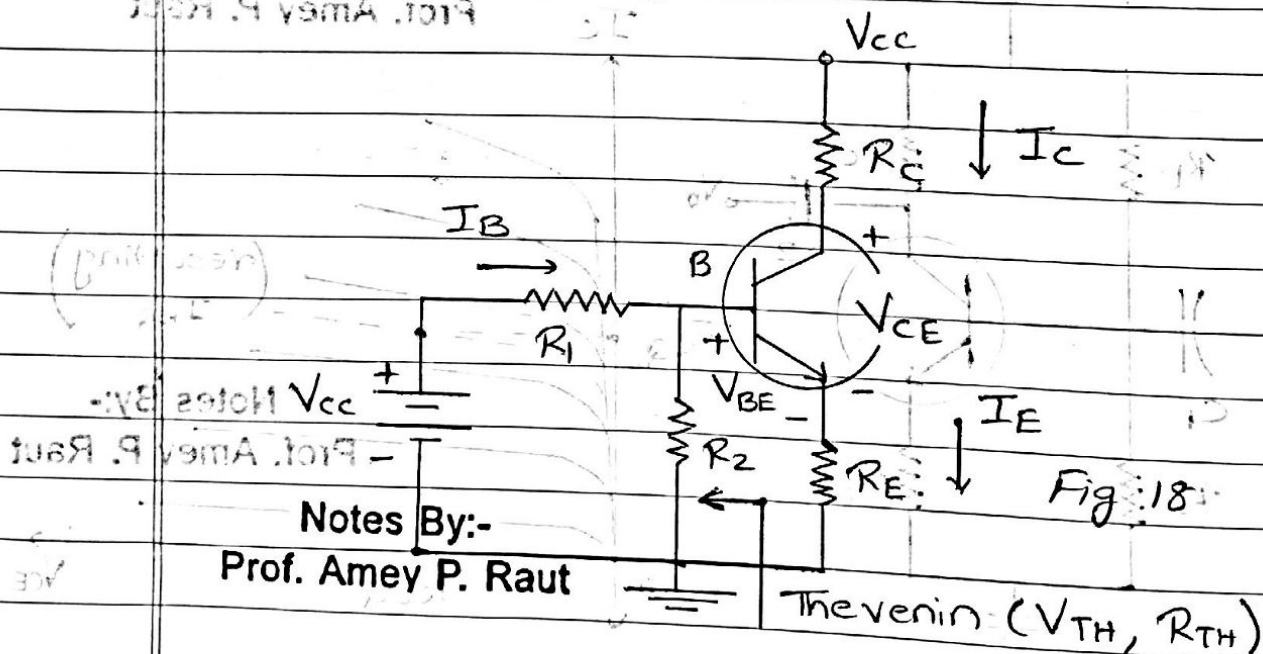
There are two methods that can be applied to analyze the voltage-divider configuration.

- i) Exact method, which can be applied to any voltage-divider configuration.
- ii) Approximate method, which can be applied only if specific conditions are satisfied.

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→ Exact Analysis



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For dc analysis the network of fig 16 can be redrawn as shown in fig 18.

The thevenin equivalent network for the input side can be found in the following manner.

$R_{TH} \rightarrow$ the voltage source is replaced by a short circuit equivalent

$$\therefore R_{TH} = R_1 \parallel R_2 \dots \dots \dots (1)$$

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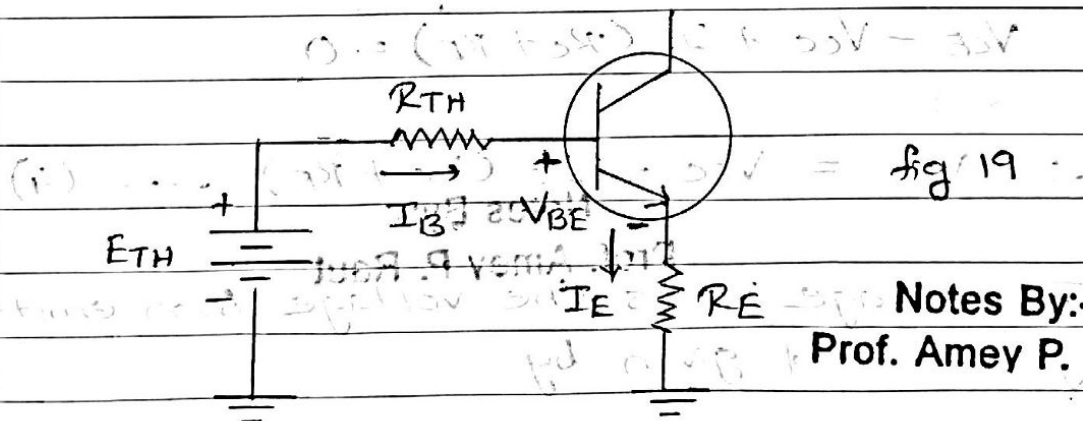
Prof. Amey P. Raut

$E_{TH} \rightarrow$ The voltage source V_{CC} is returned to the network and the open-circuit Thevenin voltage can be determined as follows.

Applying the voltage divider rule

$$E_{TH} = V_{TH} = \frac{R_2 V_{CC}}{R_1 + R_2} \dots \dots \dots (2)$$

The thevenin network can be redrawn at input side as shown in fig 19.



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I_{BQ} can be determined by applying KVL in the clockwise direction for the loop:

$$E_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0$$

Substituting $I_E = \beta(1) I_B$ and solving for I_B

$$I_B = \frac{E_{TH} - V_{BE}}{R_{TH} + (\beta + 1) R_E} \quad (3)$$

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The numerator is difference of two voltages levels and the denominator is the base resistance plus the emitter resistor reflected by $(\beta + 1)$

Applying KVL to the output loop of fig 18

$$-V_{CC} + I_E R_E + V_{CE} + I_C R_C = 0$$

Substituting $I_E \cong I_C$ and grouping terms

$$V_{CE} - V_{CC} + I_C (R_C + R_E) = 0$$

$$\therefore V_{CE} = V_{CC} - I_C (R_C + R_E) \quad (4)$$

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The voltage V_E is the voltage from emitter to ground and given by

$$V_E = I_E R_E \quad (5)$$

voltage from collector to ground can be determined from

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$$V_{CE} = V_C - V_E \quad \text{Prof. Amey P. Raut}$$

$$V_C = V_{CE} + V_E \quad \dots \dots (6)$$

$$\text{or } V_C = V_{CC} - I_C R_C \quad \dots \dots (7) \quad (\text{from 4})$$

The voltage at the base with respect to ground can be determined from

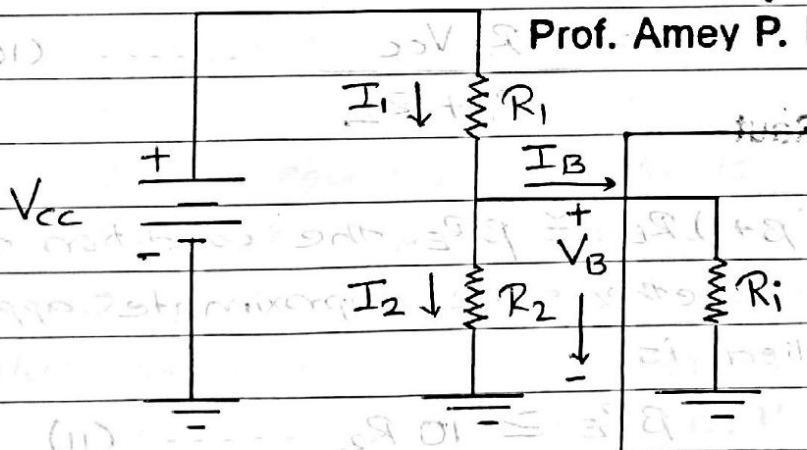
$$V_B = V_{CC} - I_B R_B \quad \dots \dots (8)$$

$$\text{or } V_B = V_{BE} + V_E \quad \dots \dots (9)$$

→ Approximate Analysis

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$$R_1 \gg R_2$$

$$(I_1 \cong I_2)$$

Fig. 20

The input section of the voltage divider configuration can be represented by the network of Fig. 20.

The resistance R_i is the equivalent resistance between base and ground for the transistor with an emitter resistor R_E .

The reflected resistance between base and emitter is defined by - Notes By:-

$$R_i = (\beta + 1) R_E \quad \text{Prof. Amey P. Raut}$$

(i) If R_i is much larger than the resistance R_2 , the current I_B will be much smaller than I_2 and I_2 will be approximately equal to I_1 .

If the approximation is assumed that I_B is essentially 0A compared to I_1 or I_2 then, $I_1 = I_2$ and R_1 and R_2 can be considered series elements.

The voltage across R_2 i.e. base voltage can be determined using the voltage divider rule, that is

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$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} \quad \dots \dots \dots (10)$$

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$$R_1 + R_2$$

$\therefore R_i = (\beta + 1) R_E \cong \beta R_E$ the condition that will define whether the approximate approach can be applied is

$$\beta R_E \geq 10 R_2 \quad \dots \dots \dots (11)$$

Thus, if β times the value of R_E is at least 10 times the value of R_2 , the approximate approach can be applied with a high degree of accuracy.

The level of V_E can be calculated from

$$V_E = V_B - V_{BE} \dots \dots (12)$$

The emitter current can be determined from

$$I_E = \frac{V_E}{R_E} \dots \dots (13)$$

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and $I_{CQ} = I_E \dots \dots (14)$

The collector - to - emitter voltage is determined by

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$I_C \cong I_E$$

$$V_{CEQ} = V_{CC} - I_C (R_C + R_E) \dots \dots (15)$$

From equations 10 to 15 it is clear that β does not appear and I_B was not calculated. The Q - point is therefore independent of the value of β .

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→ SATURATION LEVEL

The collector saturation level or maximum collector current for an voltage divider bias design can be determined by applying a short circuit between the collector - emitter terminals as shown in fig 21

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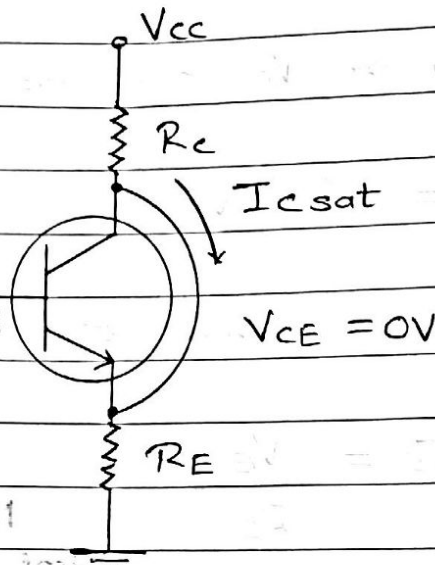


Fig 21

The resulting collector current is given by

$$I_{csat} = \frac{V_{cc}}{R_c + R_E} \quad (16)$$

The emitter resistor reduces the collector saturation level below that obtained with a fixed bias configuration using the same collector resistor.

→ LOAD LINE ANALYSIS

The collector-emitter loop is given by

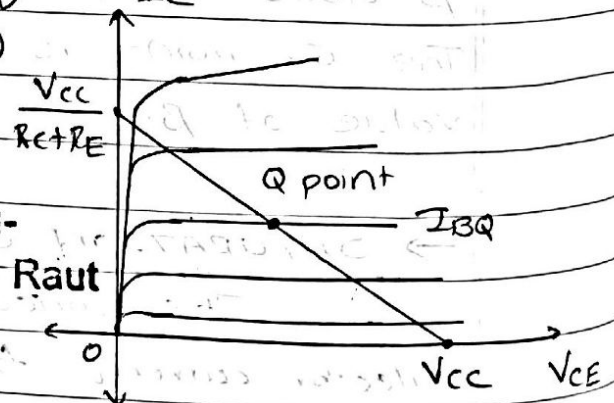
$$V_{CE} = V_{cc} - I_C (R_c + R_E)$$

From above if $I_C \neq 0A$,

$$V_{CE} = V_{cc} \quad | \quad I_C = 0A$$

and if $V_{CE} = 0V$

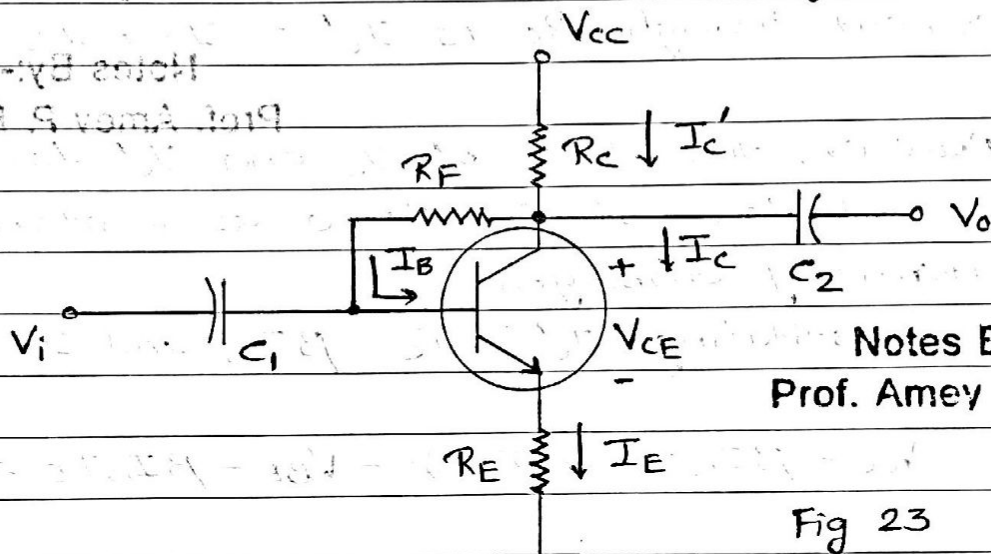
$$I_C = \frac{V_{cc}}{R_c + R_E} \quad | \quad V_{CE} = 0V$$



The value of I_{BQ} is determined by different equation.

* COLLECTOR FEEDBACK CONFIGURATION (Collector Bias)

An improved level of stability can be obtained by introducing a feedback path from collector to base as shown in fig 23.



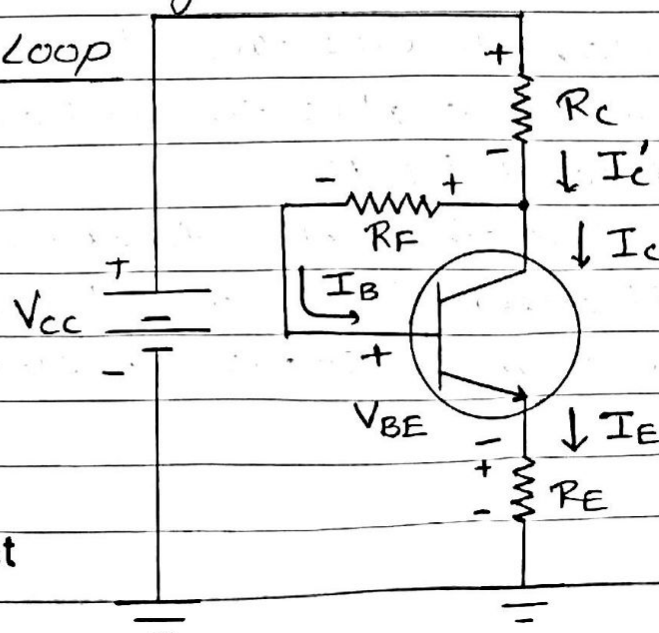
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Fig 23

Although the Q-point is not totally independent of beta; the sensitivity to changes in β to temperature variations is normally less than for the fixed bias configuration.

→ Base Emitter Loop

Fig 24



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Fig 24 shows the base-emitter loop for the voltage feedback configuration.

Applying KVL to the loop in clockwise direction we get,

$$V_{CC} - I_C' R_C - I_B R_F - V_{BE} - I_E R_E = 0$$

current through R_C is $I_C' = I_C + I_B$

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However, the level of I_C and I_C' far exceeds the usual level of I_B . Hence assumption $I_C' \cong I_C$ is normally employed.

Substituting $I_C' \cong I_C = \beta I_B$ and $I_E \cong I_C$

$$V_{CC} - \beta I_B R_C - I_B R_F - V_{BE} - \beta I_B R_E = 0$$

$$V_{CC} - V_{BE} - I_B (\beta (R_C + R_E) + R_F) = 0$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta (R_C + R_E)} \quad \text{..... (1)}$$

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Numerator is the difference of voltage levels, whereas the denominator is the base resistance plus the collector and emitter resistance reflected by β .

Therefore, the feedback path results in a reflection of the resistance R_C back to the input circuit, much like the reflection of R_E .

The equation for I_B has the following format, which can be compared with the result for the fixed-bias configuration.

$$I_B = \frac{V'}{R_F + \beta R'}$$

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For the fixed-bias configuration $\beta R'$ does not exist.

Because $I_C = \beta I_B$

$$I_{CQ} = \beta V' / (R_F + \beta R') = \frac{\beta V'}{R_F/\beta + R'}$$

In general $R' \gg R_F/\beta$

$$I_{CQ} \cong \frac{V'}{R'}$$

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The result is an equation absent of β , which would be very stable for variations in β . As R' is larger for the collector feedback configuration the sensitivity to variations in β is less.

Since R' is 0Ω for the fixed bias config. and is therefore quite sensitive to variation in β .

→ Collector - Emitter Loop

Applying KVL to collector-emitter loop of fig 23

$$I_E R_E + V_{CE} + I_C' R_C - V_{CC} = 0$$

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$I_E \approx I_C$ and $I_C' \approx I_C$ Prof. Amey P. Raut

$$I_C (R_C + R_E) + V_{CE} - V_{CC} = 0$$

$$\therefore V_{CE} = V_{CC} - I_C (R_C + R_E) \quad \dots \dots (2)$$

→ Saturation Condition

Using the approximation $I_C' \approx I_C$ we find that the equation for the saturation current is,

$$I_{Csat} = I_{Cmax} = \frac{V_{CC}}{R_C + R_E} \quad \dots \dots (3)$$

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→ Load Line Analysis

Continuing with the approximation $I_C' \approx I_C$ results in the same load line defined for the voltage-divider configuration.

* ADVANTAGES AND LIMITATIONS

→ Fixed Biasing

• Advantages:-

- 1) Simple circuit
- 2) Biasing conditions can be easily set.

• Disadvantages:-

- 1) Poor thermal stabilization

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→ Voltage Divider Biasing

• Advantages:-

- 1) Provides excellent stability
- 2) It acts as a constant current source biasing.
- 3) It provides stability for I_{CO} and β .

• Limitations:-

- 1) Biasing conditions cannot be easily obtained.

→ Collector to base biasing

• Advantages:-

- 1) Simple circuit
- 2) Biasing condition can be easily set.
- 3) Good stability compared to fixed biasing

• Limitations:-

- 1) Does not provide excellent stability.

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* BIAS STABILIZATION

The stability of a system is a measure of the sensitivity of a network to variations in its parameters. In any amplifier employing a transistor the collector current I_C is sensitive to each of the following parameters.

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- 1) β_{AC} increases with increase in temperature
- 2) V_{BE} decreases about 2.5 mV per degree Celsius ($^{\circ}\text{C}$) increase in temperature
- 3) I_{CO} :- doubles in value for every 10°C increase in temperature.

Any or all of these factors can cause the bias point to drift from the designed point of operation.

The effect of changes in leakage current (I_{CO}) and current gain (β) on the dc bias point is demonstrated by characteristics of transistor shown in fig 25 & 26 below,

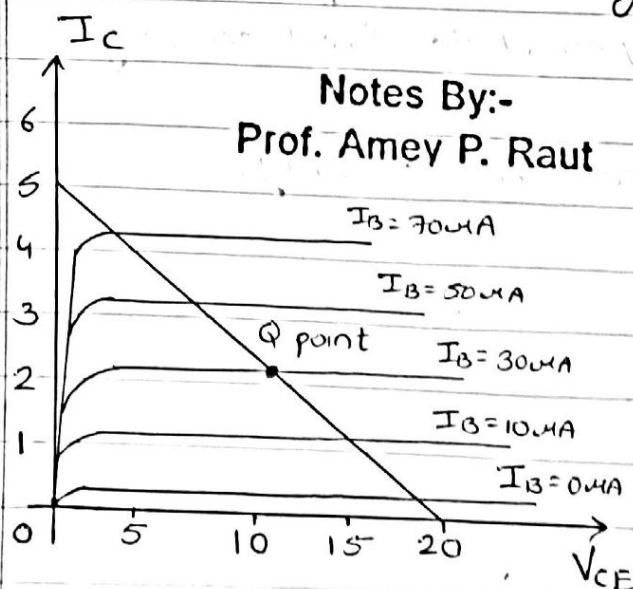


Fig 25

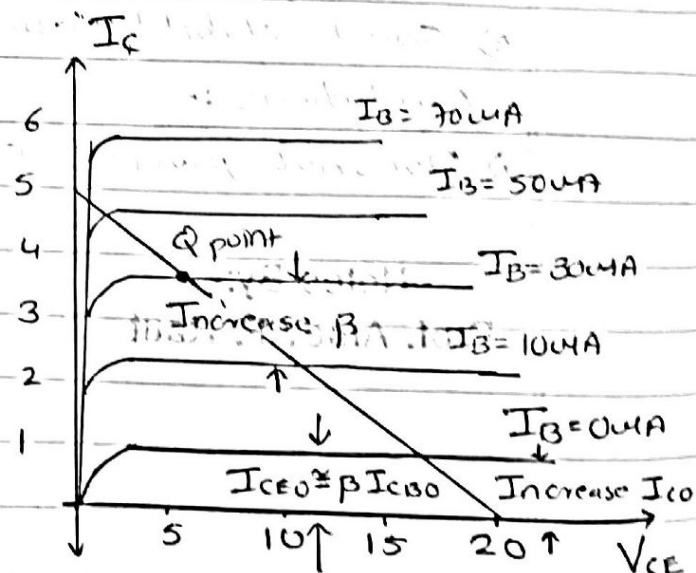


Fig 26

An arbitrary Q-point is marked in fig 25 at $I_B = 30 \mu A$. Because the fixed bias circuit provides a base current which is not affected by temperature or the change in leakage current or β , the same current magnitude will exist at high temperatures as indicated in fig 26.

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As the figure shows, this will result in the dc bias point shifting to a higher collector current and a lower collector-emitter voltage operating point.

In extreme, the transistor could be driven into saturation.

The new operating point may not be at all satisfactory, and considerable distortions may result because of the bias point shift.

A better bias circuit is one that will stabilize or maintain the dc bias initially set, so that the amplifier can be used in changing temperature environment.

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* STABILITY FACTORS $S(I_{CO})$, $S(V_{BE})$ & $S(\beta)$

A stability factor S is defined for each of the parameters affecting bias stability as follows,

$$S(I_{CO}) = \frac{\Delta I_C}{\Delta I_{CO}} \quad \text{--- (1)}$$

$$S(V_{BE}) = \frac{\Delta I_C}{\Delta V_{BE}} \quad \text{--- (2)}$$

$$S(\beta) = \frac{\Delta I_C}{\Delta \beta} \quad \text{--- (3)}$$

In each case, the delta symbol (Δ) signifies change in that quantity. The numerator of each equation is the change in collector current as established by the change in the quantity in the denominator.

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If a change in I_{CO} fails to produce a significant change in I_C , the stability factor defined by $S(I_{CO})$ will be quite small.

Networks that are quite stable and relatively insensitive to temperature variations have low stability factors. The higher the stability factor the more sensitive is the network to variations in that parameter.

→ Fixed bias configuration

$$S_{I_{CO}} \cong \beta$$

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$$S(V_{BE}) \cong \frac{-\beta}{R_B}$$

R_B

$$S(\beta) = \frac{I_{C1}}{\beta_1}$$

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Note: The notation I_{C1} and β_1 is used to define their values under one set of network conditions.

→ VOLTAGE DIVIDER BIAS

$$S(I_{CQ}) = \frac{\beta(1 + R_{TH}/R_E)}{\beta + R_{TH}/R_E}$$

In above equation $S(I_{CQ})$ has its lowest level and the network has its greatest stability when $R_E > R_{TH}$ or R_{TH}/R_E is as small as possible.

For the voltage divider bias R_{TH} is much less than the emitter resistance and still can have an effective design

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$$S(V_{BE}) = \frac{-\beta/R_E}{\beta + R_{TH}/R_E}$$

Substituting $\beta \gg R_{TH}/R_E$ we get

$$S(V_{BE}) = -\beta/R_E = -1$$

$$(\beta + R_{TH}/R_E) \beta R_E = (\beta R_E)$$

which shows that the larger the resistance R_E , the lower is the stability factor and the more stable is system.

$$S(\beta) = \frac{I_{C1}(1 + R_{TH}/R_E)}{\beta_1(\beta_2 + R_{TH}/R_E)}$$

$$\text{Notes By:- } \beta_1(\beta_2 + R_{TH}/R_E)$$

Note: * Prof. Amey P. Raut

The notation β_2 is used to define the new value of β as established by such causes as temperature change, variation in β or change in β transistor

→ Collector - Base Bias configuration

$$S(I_{CQ}) \cong \frac{\beta(1 + R_B/R_C)}{\beta + R_B/R_C}$$

if R_B/R_C is small $S(I_{CQ}) \cong 1$ thus network will be stable.

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$$S(V_{BE}) = \frac{-\beta/R_C}{\beta + R_B/R_C}$$

Substituting $\beta \gg R_B/R_C$ we get

$$S(V_{BE}) \cong -\frac{1}{R_C}$$

thus, larger the R_C , lower is the stability factor and the more stable is system.

$$S(\beta) = \frac{I_{C1}(R_B + R_C)}{\beta_1(R_B + \beta_2 R_C)}$$

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* TRANSISTOR As A SWITCH

For switching applications transistor is biased to operate in the saturation or cut off regions.

A transistor can be employed as a switch by forming a network as shown in fig 27

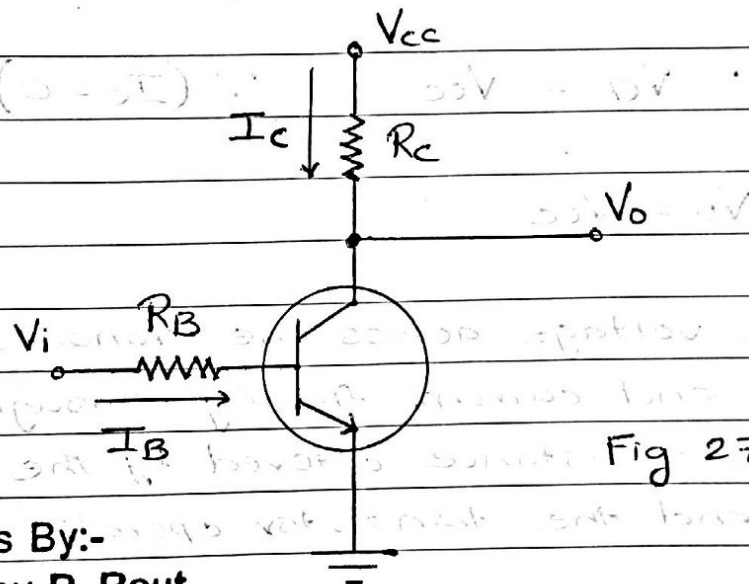


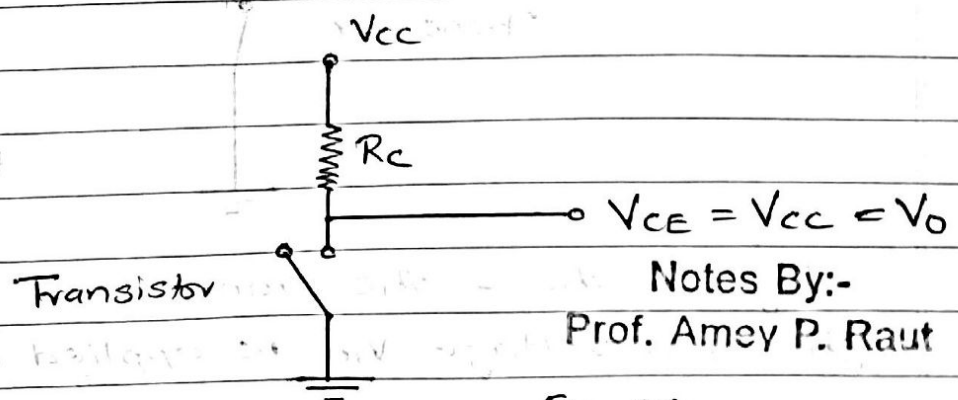
Fig 27

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Mode I :- Transistor as open switch (cut off)

In cut off region both the junctions of a transistor are reverse biased and very small reverse current flows through the transistor.



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Fig 28

The voltage drop across the transistor (V_{CE}) is high. This in the cut off region the transistor is equivalent to an open switch.

The input voltage is zero so $I_B = 0$ & $I_C = 0$ we know for collector circuit

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$$V_{CE} = V_{CC} - I_C R_C \quad \text{Prof. Amey P. Raut}$$

$$\therefore V_{CE} = V_{CC} \quad \therefore (I_C = 0)$$

$$\therefore V_O = V_{CC}$$

The voltage across the transistor is $V_{CE} = V_{CC}$ i.e high and current flowing through it $I_C = 0$, thus the resistance offered by the transistor is infinite and the transistor operates as open switch.

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MODE II:- Transistor as closed switch (saturation)

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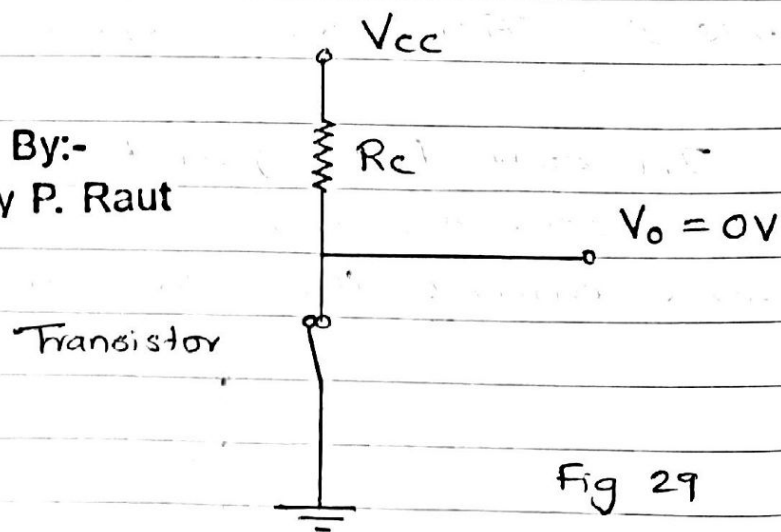


Fig 29

Fig 29 shows the transistor in saturation. A high input voltage V_{in} is applied at the base.

The value of V_{in} and resistance R_B are adjusted such that a large I_B flows. This will saturate the transistor. The value of I_B should be such that the following condition should be satisfied

$$I_B \geq \frac{I_{csat}}{\beta}$$

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In the saturation region both the junctions of a transistor are forward biased. The voltage drop across the transistor is very small thus

$$V_o \approx 0V$$

A large current flowing through transistor indicates that it is equivalent to closed switch

Output waveform

